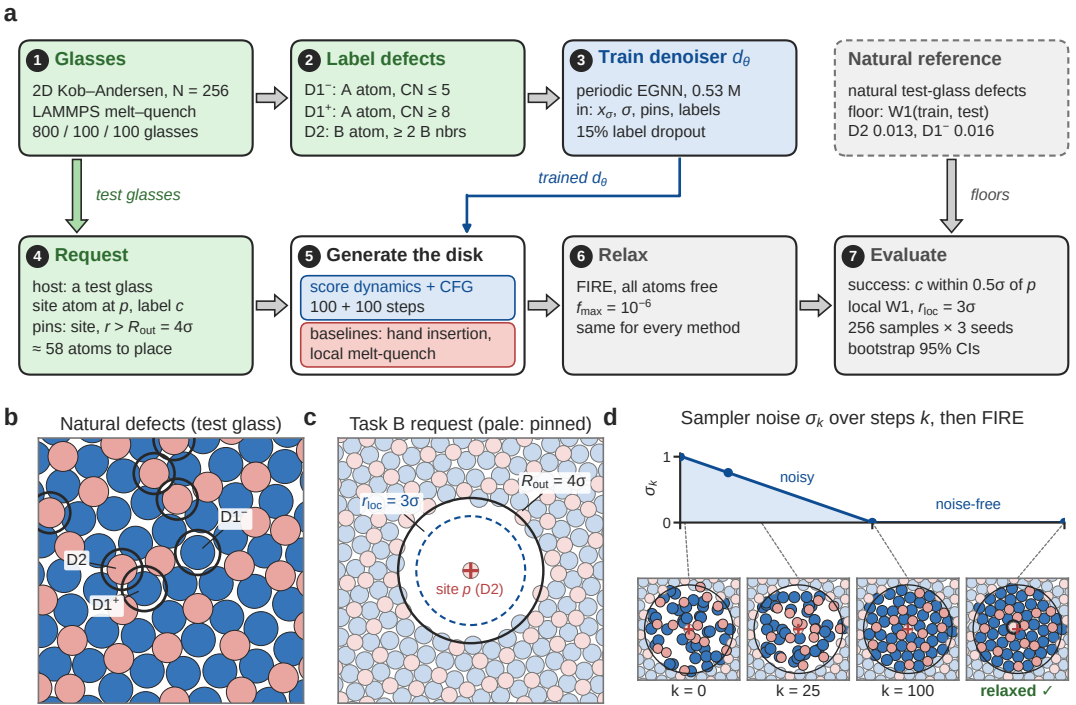

a
1 Glasses
2D Kob–Andersen, N = 256
LAMMPS melt–quench
800 / 100 / 100 glasses
2 Label defects
D1⁻: A atom, CN ≤ 5
D1⁺: A atom, CN ≥ 8
D2: B atom, ≥ 2 B nbrs
3 Train denoiser d_θ
periodic EGNN, 0.53 M
in: x_σ, σ, pins, labels
15% label dropout
Natural reference
natural test-glass defects
floor: W1(train, test)
D2 0.013, D1⁻ 0.016
test glasses
trained d_θ
floors
4 Request
host: a test glass
site atom at p, label c
pins: site, $r > R_{out} = 4\sigma$
≈ 58 atoms to place
5 Generate the disk
score dynamics + CFG
100 + 100 steps
baselines: hand insertion,
local melt-quench
6 Relax
FIRE, all atoms free
$f_{max} = 10^{-6}$
same for every method
7 Evaluate
success: c within 0.5σ of p
local W1, $r_{loc} = 3\sigma$
256 samples × 3 seeds
bootstrap 95% CIs
b
Natural defects (test glass)
D1⁻
D2
D1⁺
c
Task B request (pale: pinned)
$r_{loc} = 3\sigma$
$R_{out} = 4\sigma$
site p (D2)
d
Sampler noise σ_k over steps k, then FIRE
1
0
σ_k
noisy
noise-free
k = 0
k = 25
k = 100
relaxed ✓